

1 Experimental Design

1.1 Objective

This simulation study evaluates the performance of the automatic noise constant selection procedure, denoted $\hat{k}$, within the matrix-variate heterogeneous clustering (HC) framework implemented via `matrix_variate_noise_fit`. The primary goal is to assess:

- the accuracy and stability of the selected noise scale parameter $\hat{k}$,
- the ability of the method to recover latent cluster structure,
- robustness under multiple contamination regimes and increasing data complexity.

Two initialization strategies are compared: K-means initialization and ECME initialization.

1.2 Data-Generating Process

1.2.1 Matrix-variate mixture model

Each observation is a matrix $X_i \in \mathbb{R}^{r \times p}$ generated from a finite mixture of matrix-variate Gaussian distributions:

$$X_i \sim \sum_{g=1}^G \pi_g \mathcal{MN}(M_g, U_g, V_g),$$

where $M_g \in \mathbb{R}^{r \times p}$ denotes the component mean matrix, $U_g \in \mathbb{R}^{r \times r}$ the row covariance matrix, $V_g \in \mathbb{R}^{p \times p}$ the column covariance matrix, and π_g the mixture proportions.

Covariance matrices are randomly generated to ensure positive definiteness and heterogeneity in cluster structure.

1.2.2 Dimension settings

The simulation considers a range of matrix dimensions:

$$(r, p) \in \{(2, 2), (2, 4), (3, 3), (3, 5), (4, 4), (5, 5), (6, 6)\}.$$

This allows evaluation across low-, moderate-, and higher-dimensional regimes.

1.2.3 Sample sizes and mixture complexity

Two sample sizes are considered:

$$n \in \{150, 300\},$$

and the number of mixture components is set to:

$$G \in \{2, 3\}.$$

1.3 Contamination Mechanisms

To assess robustness, four contamination mechanisms are introduced.

1.3.1 Matrix-level contamination

Entire observations are replaced with noise:

$$X_i \sim U(a, b)^{r \times p}.$$

1.3.2 Column-wise contamination

A randomly selected column of selected matrices is replaced with uniform noise, inducing structured partial corruption.

1.3.3 Element-wise contamination

Individual entries are corrupted according to a Bernoulli mask:

$$X_{ij} \leftarrow \begin{cases} U(a, b), & \text{with probability } \rho, \\ X_{ij}, & \text{otherwise.} \end{cases}$$

1.3.4 Permutation contamination

Selected matrices undergo random permutation of their entries, preserving marginal distributions but destroying structural dependence.

1.3.5 Contamination levels

The contamination proportion is varied as:

$$\rho \in \{0.00, 0.05, 0.10, 0.20, 0.30, 0.40\}.$$

1.4 Monte Carlo Design

For each configuration, $R = 100$ Monte Carlo replications are generated. The full factorial design spans:

- matrix dimension (r, p) ,
- sample size n ,
- number of components G ,
- initialization method,
- contamination mechanism,
- contamination level.

1.5 Model Fitting Procedure

For each simulated dataset, the HC model is fitted using:

```
matrix_variate_noise_fit.
```

Two initialization strategies are considered:

- K-means initialization,
- ECME initialization.

The noise constant is selected automatically via:

```
estimate_k = TRUE.
```

The fitted model returns:

- cluster assignments,
- estimated noise proportion,
- selected noise constant $\hat{k}$.

1.6 Evaluation Metrics

1.6.1 Clustering performance

Two standard metrics are used:

- Misclassification Rate:

$$\frac{1}{n} \sum_{i=1}^n \mathbb{I}(z_i \neq \hat{z}_i),$$

- Adjusted Rand Index (ARI), computed using `mclust`.

1.6.2 Noise estimation performance

The primary object of interest is the selected noise constant:

$$\hat{k} = \text{estimate_k}(X).$$

Performance is assessed through:

- stability of $\hat{k}$ across replications,
- sensitivity to contamination level,
- sensitivity to initialization method.

1.7 Output Structure

Two datasets are produced:

1.7.1 Summary output

`estimate_k_summary.csv` contains one row per experiment, including:

- dimension, sample size, number of components,
- initialization method,
- contamination type and level,
- estimated $\hat{k}$,
- ARI and misclassification rate,
- number of contaminated observations.

1.7.2 Candidate-level log

`estimate_k_candidates.csv` contains diagnostic-level outputs, including:

- full cluster assignment tables,
- replicate-level configuration metadata,
- initialization and contamination settings.

1.8 Key Comparisons

The study evaluates:

- sensitivity of $\hat{k}$ to initialization strategy,
- robustness under increasing contamination,
- scalability across matrix dimensions,
- identifiability and stability of the noise constant.

1.9 Overall Aim

The central question is whether the automatic HC noise calibration procedure reliably recovers an appropriate noise scale across heterogeneous matrix structures and contamination regimes, and whether this behavior depends on the initialization strategy.